Is Pakistan Realizing its Demographic Dividend? A Time-Series Analysis of Economic Adjustment to a Changing Age Structure

Abstract: There has been considerable research on demographic dividends in regional contexts, particularly in East Asian countries. In literature, there is an emphasis on the demographic dividend that can be reaped with the implementation of appropriate policies. While most studies decompose the contribution of the demographic dividend, we extend this analysis to the context of Pakistan, delving into whether, up till now, Pakistan has been able to convert its fertility decline into meaningful economic growth through channels of savings and capital formation. We employ time-series models, including VECM and the Mediation model, to explore the relationships between demographic variables and macroeconomic ones in the case of Pakistan. Our results suggest that Pakistan's demographic dividend remains a potential rather than a realized asset, requiring targeted policy interventions beyond demographic management alone.

Keywords: Demographic Dividend, Pakistan, VECM, Mediating Model, Forecasting, Demographic Transition.

1 Introduction

Over the last few decades, Pakistan's fertility rate has declined from over 6.8 births per woman in the 1960s to around 3.1 today, resulting in an increase in the share of the working-age population. According to the demographic dividend theory, this should catalyze higher savings, investment, and sustained capital growth. Yet, Pakistan's savings rate remains among the lowest in South Asia, its capital formation has stagnated significantly, and the per capita growth continues to lag behind its regional peers (World Bank, 2024). This disconnects between the demographic transition and the expected economic performance presents a puzzle, which we aim to untangle by examining whether an increase in the working-age population translates into a long-term dividend through higher savings and capital accumulation.

The demographic dividend framework posits a growth opportunity where the first, transitory dividend arises from a larger labor force and the second, sustained dividend is driven by increased savings and capital accumulation as families have fewer children (Bongaarts & Hodgson, 2022). Although the demographic dividend is well-documented in East Asia and elsewhere, its applicability to Pakistan is still inadequately tested. As Durr-e-Nayab has warned, without the right conditions, this youth bulge will not automatically become a dividend; rather, it may even convert into a "Demographic Threat" (2008).

2 Literature Review and Motivation

Older debates on population growth have been characterized by a contrast between those who believe that population growth hinders economic growth, such as Malthus, and those who associate population growth with economic development, including Kuznets. Research over the past few decades has highlighted a more nuanced approach, emphasizing the significance of demographic transition and the growth of working-age populations. Simply, the 'demographic dividend' is realized when the working-age population increases as a proportion of the population, thereby creating opportunities for savings, productivity and economic growth.

Barro established a negative association between fertility and per capita economic growth, using cross-country regressions, whereby growth was measured as the average annual growth rate, to explain why some countries grew faster than others (1991). Drawing on the Life-Cycle Hypothesis, which states that individuals save during their working years and dissave when they are young or old, hence the aggregate savings of an economy depend on the age structure of the population, Bloom and Williamson argue that fertility decline contributes to economic growth.

Using panel data of Asian countries, they attribute part of East Asia's growth miracle to a rise in the working-age population. They treat demographic transition as exogenous, an approach that we also adopt (Bloom and Williamson, 1998). Bloom, Canning, and Malaney, who identify the channels through which demographic change translates into economic growth, namely labour supply, savings, and human capital investment and also find that South Asia is a late-transition region with an emerging dividend (1998).

Literature has shown that East Asian countries, such as South Korea, succeeded because they had strong policies in place to convert their growing working-age population into real economic gains (Bloom & Williamson, 1998). In Pakistan, Dur-e-Nayaab highlights the demographic dividend as a time-bound opportunity, noting that the gains from it depend on supportive policies (2006). She also notes that Pakistan is currently in the midst of a critical demographic window (1990-2045). However, her study relies on descriptive statistics and lacks rigorous econometric testing of these proposed linkages.

We develop our conceptual framework underlying our models through these seminal papers, as well as theoretical insights provided by Bongarts and Hodgson, who describe a demographic dividend arising from demographic changes, specifically a rise in the working-age population, shaping higher savings and investment, which become long-run channels for economic growth (Bongaarts & Hodgson, 2022).

Interestingly, most of the current literature looks at growth models, and on regions as a whole. We distinguish from past research, by focusing singularly on Pakistan and using a time-series analysis to research whether the relationship between these variables hold in the first place and if so, how significant is their association.

To explore this theory in Pakistan's context, we aim to answer: How has Pakistan's demographic transition shaped long-run and short-run dynamics of domestic savings, capital formation, and economic growth? We use a VECM model to examine the long run equilibrium relationship between savings, capital formation and economic growth in Pakistan, conditioning on the country's demographic transition. Our mediation model aims to capture a link between the working age population to GDP growth, mediated by capital formation. We also supplement this by employing VAR and VECM for forecasting to explore the future trajectory of these dynamics over the timespan the country's demographic window remains open.

3 Data

3.1 Data Source and Variables

We utilize annual time series data for Pakistan spanning 1990 to 2023 from the World Bank's Development Indicators database. We use demographic transition and economic performance and convert each into a balanced time-series, where periodicity of the data is annual. Table 1 outlines the variables, and the justification for including these variables based on literature.

Variable Name	Description	Justification for Inclusion
Fertility Rate (SP.DYN.TFRT.IN)	Total fertility rate (births per woman)	Bloom & Williamson (1998) Bloom, Canning & Malaney (2001) Bongaarts & Hodgson (2022) Durr-e-Nayab (2008)
Domestic Savings (NY.GDS.TOTL.ZS)	Gross domestic savings (% of GDP)	Bongaarts & Hodgson (2022) Durr-e-Nayab (2008)
Capital Formation (NE.GDI.TOTL.ZS)	Gross capital formation (% of GDP)	Bongaarts & Hodgson (2022)
Working Age Population (SP.POP.1564.TO.ZS)	Population ages 15-64 (% of total population)	Bloom & Williamson (1998) Bloom, Canning & Malaney (2001) Bongaarts & Hodgson (2022) Durr-e-Nayab (2008)
Dependency Ratio (SP.POP.DPND)	Age dependency ratio (% of working age population)	Durr-e-Nayab (2008)
GDP per Capita (NY.GDP.PCAP.KD)	GDP per capita in current US dollars	Bloom, Canning & Malaney (2001) Bongaarts & Hodgson (2022) Durr-e-Nayab (2008)

Table 1: Variable Definitions, Definition, and Justification.

3.2 Structural Break in Fertility Data

We restricted our sample to the post-1990 period based on a permanent structural break in fertility's long-term trend driven by policy reforms, family planning programs, improved female education, and urbanization (Soomro 2000). This is also confirmed by our preliminary visual and statistical analysis of Pakistan's fertility rate trajectory, as the fertility rate has not returned to its previous path. Since Pakistan's demographic transition window also began in 1990, this year marks the start of structural change, allowing us to get more reliable estimates (Dur-e-Nayaab,2006).

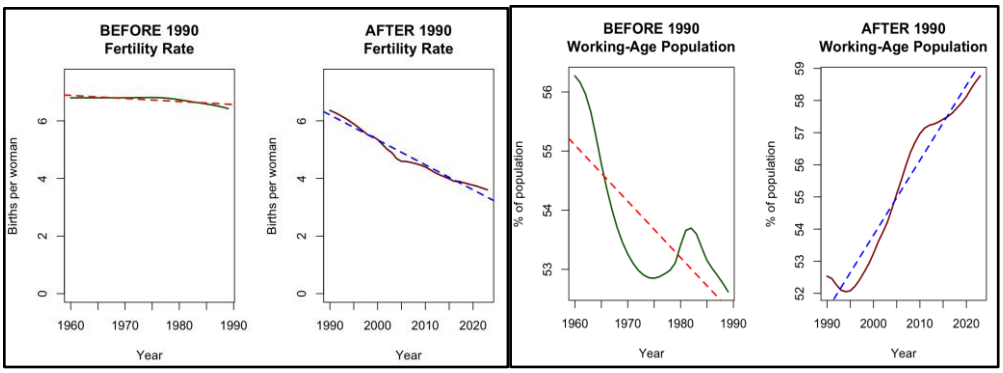


Figure 1: Graphs 1 & 2: Fertility and Working Age Population (before and after 1990).

3.3 Descriptive Statistics and Trends

Autoplots for all demographic and macroeconomic variables were constructed (Appendix: Figure A1), and summary statistics are also presented.

Variable	Mean	SD	Min	Max	N
Fertility	4.78	0.88	3.60	6.36	34
Domestic Savings	10.99	3.93	3.61	17.40	34
Capital Formation	16.70	1.77	13.97	20.69	34
Working Age Population	55.32	2.38	52.05	58.77	34
Dependency Ratio	81.08	7.83	70.17	92.12	34
GDP per Capita	1236.26	215.99	950.88	1642.28	34

Table 2: Variable Summary Statistics (1990-2023).

4 Methodology

4.1 Preliminaries

The Augmented Dickey-Fuller (ADF) test was used to assess the stationarity of all time series variables; however, in small sample sizes, the test exhibits low power. Excessive lags can further reduce power by a rapid decrease in degrees of freedom without substantial gains in controlling serial correlation. Therefore, to address this trade-off, a lag length of 1 was adopted (Aylar et al., 2017; Najam & Akmal, 2020). To complement the ADF test, visual inspection using time-series plots and autocorrelation function (ACF) plots were also used *(Appendix: Figure B2 and Table A1).*

The ADF results reveal that GDP per capita, domestic savings, fertility rate, and capital formation are non-stationary, whereas the working-age population and dependency ratio are stationary *(Appendix Table A1)*. The plots reveal that capital formation, savings and GDP per capita exhibit stochastic behavior, and demographic variables like fertility rate, working age

population, and the dependency ratio display a pronounced deterministic trend. Moreover, the ACF plots also show persistent autocorrelation for all time series variables, including the working-age population and the dependency ratio, despite their classification as stationary by the ADF test. Taken together, these findings suggest that our variables contain strong deterministic components and persistent dynamics that may not be fully captured by formal unit root tests in a small sample, reinforcing the importance of combining statistical tests with graphical diagnostics.

After first differencing, the macroeconomic variables became stationary, also confirmed by the ADF test. However, fertility still fails to reject the null (see Table 3).

Variable	Test Statistic	P-Value	Stationary (5%)
Δ GDP per capita	-4.110	0.0176	YES
Δ Domestic savings	-6.599	0.0100	YES
Δ Capital formation	-3.994	0.0218	YES
Fertility (detrended)	-1.157	0.8989	NO
Working-age population (detrended)	-4.440	0.0100	YES
Dependency ratio (detrended)	-4.376	0.0100	YES

Table 3: ADF results for Variables (after first-difference/detrending).

Note: Stationary assessed at the 5% significance level.

After ACF plots indicate minimal autocorrelation for macroeconomic series, whereas demographic variables still show persistence (*Appendix: Figure B3*). After detrending, autoplots appear more stable around their means with relatively constant variance. The fertility rate exhibits periodic fluctuations, following a downward trend in the early years and an upward trend in recent years. The working-age population also displays a regular cycle, indicating long-term demographic shifts. Such smooth, long-term cycles in age structure are understood to result from enduring transitions in fertility and mortality, shaping the economic life cycle (Lee & Mason, 2010). After thorough analysis, the macroeconomic variables (GDP per capita, capital formation, and savings) are recognized as I(1), while the demographic variables (fertility, working-age population, and dependency ratio) exhibit deterministic trends *(Appendix: Figure B4).*

4.2 VECM

A Vector Error Correction Model (VECM) was implemented to empirically examine demographic dividend in Pakistan - a framework suited to distinguishing between permanent long-run relationships and transitory short-run fluctuations. We treat population structure as exogenous to a dynamic system of endogenous macroeconomic variables in accordance with literature (Bloom and Williamson, 1998; Aksoy et al. 2016). Population structure proxied by the working age population exhibits a slow-moving deterministic trend and does not need to be I(1) to be incorporated in the VECM model. Furthermore, Dur-e-Nayab, explains how the working age population successfully captures the potential for a demographic dividend, summarizing the effects of declining fertility and dependency ratios, consistent with Pakistan's ongoing demographic transition (2008). The endogenous variables are GDP per capita, Domestic Savings, and Capital Formation, all of which are I(1) macroeconomic variables whose interconnected dynamics we wish to model.

For lag selection, AIC, HQ, and FPE suggest a lag length of 4, while the Schwarz Criterion (SC) suggests a lag length of 1. Based on the principle of parsimony to avoid overfitting our limited sample, a lag length (p) of 2 was chosen. This decision was made based on the parameter intensive nature of data and the limited sample size. Conditional on one cointegrating vector (**r = 1**), the VECM was estimated as:

$$\Delta Y_t = \alpha\beta' Y_{t-1} + \Sigma \Gamma_i \Delta Y_{t-i} + \Psi \Delta X_t + \varepsilon_t$$

where $\mathbf{Y_t}$ are endogenous I(1) variables, $\mathbf{X_t}$ exogenous variables, $\boldsymbol{\beta}$ the cointegrating vector, $\boldsymbol{\alpha}$ the speed-of-adjustment vector, $\boldsymbol{\Gamma_i}$ short-run coefficients, $\boldsymbol{\Psi}$ exogenous effects, and $\boldsymbol{\varepsilon_t}$ white noise.

4.3 Mediation Model

To really understand how demographic transitions translate into economic outcomes, particularly in Pakistan's institutional context, we must examine the intervening transmission mechanisms. Traditional growth theories, including the Harrod-Domar and Solow models emphasize the accumulation of physical capital as a fundamental driver of economic expansion (Mankiw et al., 1992; Barro and Sala-i-Martin, 2004). Building on this foundation, Jehan and Khan demonstrate that physical capital serves as a crucial mediating channel between demographic variables and economic growth in Pakistan (2019). Their study, which covers 1960-2015, reveals that while the direct impact of demographic variables on growth varies, the

indirect effects through capital formation consistently shape the total relationship. Our study extends their framework by examining the post-1990 period, corresponding to Pakistan's demographic dividend phase to see whether capital formation mediates the relationship between the working-age population and GDP per capita in Pakistan or not.

We employ the Baron and Kenny mediation framework, adapted for time series data with appropriate econometric adjustments. Under this, mediation exists if: **(1)** the independent variable [X] affects the mediator [M], **(2)** the mediator [M] affects the dependent variable [Y] and **(3)** the inclusion of the mediator reduces the direct effect of the independent variable [X] on the dependent variable (Y). Our conceptual model consists of the following steps:

1. **Total Effect (c):** This tests if the working-age population (WorkingAge) has a significant effect on GDP per capita (GDPpc): $\Delta GDPpc_t = \alpha_0 + c * WorkingAge_t + \varepsilon_t$
2. **Path a:** This tests if WorkingAge significantly affects the mediator, Capital Formation (Capital): $\Delta Capital_t = \alpha_1 + a * WorkingAge_t + \varepsilon_1_t$

   where, a is the effect of WorkingAge on Capital.

3. **Path b + Direct Effect (c'):** This tests if Capital significantly affects GDPpc after controlling for WorkingAge: $\Delta GDPpc_t = \alpha 2 + c' * WorkingAge_t + b * \Delta Capitalt + \varepsilon_2t$

   where, b is the effect of Capital on GDPpc, controlling for WorkingAge.

4. **Indirect Effect:** This tests if the product of the coefficients from Path a and Path b ($a \times b$) is statistically different from zero. If it is, and the direct effect (c') shrinks compared to the total effect (c), then mediation is present: $IE = a \times b$

We converted the variables into stationary series by taking their first differences, symbolized as ΔCapital and ΔGDPpc in the above equations. To account for the possibility of autocorrelation in our time series data, in this model we used Newey-West Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors to calculate reliable p-values for our regression coefficients. Lastly, instead of relying on the often-violated normality assumption for the indirect effect, we used a Moving Block Bootstrap with 5,000 repetitions. This method helped us preserve the time-dependent structure of our data and provide robust confidence intervals for the indirect effect ($a \times b$).

4.4 Forecasting

4.4.1 VAR Forecasting

We examine short-run dynamic interactions between variables using a Vector Autoregression (VAR) model where all variables are treated as endogenous, in accordance with standard VAR practice. We use VAR as a short-run forecasting and shock-transmission model. Since all the variables are non-stationary, we first-difference them and then use the resulting series in the VAR model, allowing each variable to be treated as jointly endogenous.

In order to comply with econometric validity and avoid spurious results, we use first differences for the estimation and the VAR Model in first differences is as followed:

$$\Delta Yt = c + \Phi 1 \Delta Yt-1 + \cdots + \Phi p \Delta Yt-p + \varepsilon t$$

For selection of the lag length, we analyze different criteria, where Akaike's information criterion (AIC), Hannan and Quinn information criterion (HQIC) and final prediction error (FPE) recommend p = 3, while Schwarz's Bayesian information criterion (BIC) favours p = 1. Given that this is a forecasting model yet also has a smaller sample size after differencing, we choose p = 2 as the optimal lag length in order to preserve degrees of freedom and avoid overfitting while still capturing demographic dynamics. This is also supported by Ivanov and Killian, who suggest parsimony in lag selection of smaller samples by using BIC (2005).

$$\Delta Yt = c + \Phi 1 \Delta Yt-1 + \Phi 2 \Delta Yt-2 + \varepsilon t$$

We also conduct a model stability test using the roots of the characteristic polynomial. This is to ensure that demographic and macroeconomic shocks are not explosive in nature, and the VAR system is stable. The eigenvalues must lie inside the unit circle (modulus < 1). All eigenvalues lie inside the unit circle, thereby confirming that VAR (2) is dynamically stable *(Appendix Figure B5)*. We also recursively order demographic variables before macroeconomic variables, and an Impulse Response Function (IRF) is used to trace the dynamic effects of fertility shocks on macroeconomic variables in first differences. Forecast Error Variance Decomposition (FEVD) is employed to assess the relative contributions of other variables' shocks and those from its own past values to fluctuations in GDP per capita.

For our forecasting model, we convert the predictions originally in differences to that of levels, and use that to construct graphs for all variables in the system using a confidence interval of 95%. A forecast horizon of 20 years is chosen, which amounts close to the year 2045, which

literature deems as the year in which the demographic transition window for Pakistan closes (Dur-e-Nayaab 2006).

4.4.2 VECM Forecasting

We also utilize the Vector Error Correction (VECM) Model to predict future values of the domestic savings, capital formation, and GDP per capita growth, which are taken as endogenous variables. The VECM model adds value by enforcing a long-run equilibrium between key economic indicators, ensuring that forecasts do not diverge unrealistically over time. To forecast, we first determine the optimal lag length of five using the Akaike Information Criterion (AIC). The AIC is preferred over alternative criteria as it performs relatively well in small samples and reduces the risk of underfitting dynamic relationships.

The Johansen cointegration test confirms a stable long-run relationship among the endogenous variables, supporting estimation of a VECM with one cointegrating vector and exogenous demographic variables. This ensures deviations from equilibrium are corrected over time, an advantage over VAR models when cointegration exists. We forecast up to 2043, projecting exogenous variables separately: the working-age population using an ARIMA model, and fertility based on recent levels with modest stochastic variation. These are fed into the VECM to produce conditional forecasts for the endogenous variables. Overall, the VECM forecasting complements the VAR results by providing structurally informed, long-run projections, making it suitable for policymakers.

Findings & Discussion

5.1 VECM

The Johansen cointegration test (trace statistic) was then applied to the system of endogenous variables to test for the existence of a long-run equilibrium relationship. The eigenvalues indicate the strength of cointegration: $\lambda_1 = 0.529$ (strong), $\lambda_2 = 0.196$ (moderate), $\lambda_3 = 0.111$ (weak), and $\lambda_4 \approx 0$ (no cointegration). The largest eigenvalue here is 0.529, this indicates a relatively strong long-run equilibrium relationship among the variables.

Hypothesis	Test Statistic	10% Critical Value	5% Critical Value	Inference
$r = 0$	34.81	32.00	34.91	Reject H_0 at 10%
$r \leq 1$	10.75	17.85	19.96	Cannot Reject H_0
$r \leq 2$	3.78	7.52	9.24	Cannot Reject H_0

Table 4: Johansen Cointegration Test Results (Trace Statistic)

The test rejects the null of no cointegration at the 10% significance level (The trace statistic (34.81) exceeds the 10% critical value (32.00) but falls marginally below the 5% critical value (34.91). This provides evidence for the existence of a single long-run cointegrating relationship among GDP per capita, domestic savings, and capital formation in Pakistan.

The VECM model provides empirical results that confirm and validate the theoretical channels of demographic transition in Pakistan. The cointegrating vector represents a stable long-run equilibrium condition linking GDP per capita, domestic savings, and capital formation, rather than causal elasticities. The estimated long-run equilibrium (cointegrating) relationship is (*Appendix Table A2):*

$$ECT_t = 1.000 \cdot gdp_pct + 207.927 \cdot dom_sav_t + 19.695 \cdot cap_form_t - 18307.508$$

$$ECT_t = gdp_pc_t - (-207.927 \cdot dom_sav_t - 19.695 \cdot cap_form_t + 18307.508)$$

Estimated Short-Run VECM Equations are below *(Appendix Table A4)*:

$$\Delta gdp_pct_t = 0.098 \cdot ECT_{t-1} + 26.021 \cdot \Delta working_age_t + 15.414 \cdot \Delta dom_sav_{t-1} + \varepsilon_t$$ *(Standard Errors: 0.035, 9.068, 4.766)*

$$\Delta dom_sav_t = -0.006 \cdot ECT_{t-1} - 1.584 \cdot \Delta working_age_t - 1.086 \cdot \Delta dom_sav_{t-1} + \varepsilon_t$$ *(Standard Errors: 0.001, 0.371, 0.195)*

$$\Delta cap_form_t = 0.002 \cdot ECT_{t-1} + 0.517 \cdot \Delta working_age_t + 0.151 \cdot \Delta dom_sav_{t-1} + \varepsilon_t$$ *(Standard Errors: 0.001, 0.318, 0.167)*

Note: *Equations include all relevant lagged differenced terms as per the selected lag length (**p = 2**). Only key coefficients are shown for clarity.*

The working-age population, included as an exogenous variable, has a statistically significant short-run association on GDP per capita and domestic . The findings confirm the Life-Cycle Hypothesis framework outlined by Bloom et al. (1998) , as evidenced by the positive sign on GDP per capita (26.02, $p<0.01$) and the negative sign on savings growth (-1.584, $p<0.01$), where a larger labor force coincides with increased output and reduced savings.

There is a significant positive impact of changes in savings on GDP per capita (15.414, $p < 0.01$).This highlights the important role that savings plays as a financing mechanism to achieve GDP growth (Bongaarts & Hodgson). The important ECTs for GDP (positive) and savings (negative) indicate a stable system that is self-correcting for errors in its steady-state path.The adjustment coefficients (α) indicate that GDP per capita adjusts fastest to disequilibrium (0.098), followed by domestic savings (-0.006), while capital formation shows no significant adjustment (0.002, $p=0.109$).

The VECM analysis indicates that short-term changes in Pakistan's demographic structure, when included as an exogenous variable, are associated with variations in GDP per capita and domestic savings. These findings suggest that demographic factors may play a significant role in shaping output and the savings-investment channel. This aligns with the demographic dividend framework, described as a window of opportunity, although Pakistan has yet to fully realize it due to weak short-run capital formation and an insignificant error-correction response (Dur-e Nayab, 2008). The result suggests structural constraints in the investment channel, which may limit the economy's ability to fully capitalize on demographic changes in the short term for sustained growth. The significant eigenvalue and adjustment coefficients suggest that policy interventions should focus on strengthening the capital formation channel to fully realize Pakistan's demographic dividend potential. The VECM analysis provides support for the notion that Pakistan is falling short of harnessing the economic benefits of the demographic dividend. The existence of a stable cointegrating relationship leads us to calculate an impulse response analysis and forecast variance decomposition in our VAR model.

Post-estimation diagnostics indicate that the VECM is well-specified. For GDP per capita ($DW = 2.07$, $p = 0.56$), domestic savings ($DW = 1.86$, $p = 0.33$), and capital formation ($DW = 1.91$, $p = 0.39$), the Durbin-Watson statistics are close to 2, with insignificant p-values, indicating no residual correlation. Bruesch-Pagan test results show no evidence of heteroskedasticity across all equations ($p > 0.32$), confirming constant variance.

5.2 Mediation Model

In **path a,** we observe that the relationship between the working-age population share and capital formation is statistically insignificant (coefficient ~0, p = 0.98). This is the critical breakdown in the hypothesized pathway. However, **path b** shows a strong and clear relationship, whereby a 1 percentage point increase in capital formation (as a share of GDP) is associated with an approximately $16.09 increase in annual GDP per capita growth. This confirms that investment is a powerful driver of economic growth.

Effect	Path	Coefficient	P-value	95% Bootstrap CI	Interpretation
Total Effect (c)	X → Y	2.13	0.223	-	Not Significant
Path a	X → M	0.0013	0.982	-	Not Significant
Path b	M → Y	16.09	<0.001	-	**Highly Significant**
Direct Effect (c')	X → Y (controlling for M)	2.11	0.056	-	**Marginally Significant**
Indirect Effect (a×b)	X → M → Y	0.0208		[-2.97, 1.75]	Not Significant

Table 5: Results from the Mediation model.

The **indirect effect is** also negligible and insignificant (See Appendix: Figure B6). The bootstraps distribution plot below shows that the indirect effect estimate is centered at zero, with its 95% confidence interval spanning both negative and positive values. This visual evidence strongly confirms that mediation through capital formation is not statistically significant. Moreover, the **direct effect** of the working-age population on GDP is somewhat suggestive as it is positive and marginally significant (p=0.056). This hints that demographics might influence growth through other channels such as labor supply, productivity or human capital. Our diagnostic checks confirm that our models are well-specified *(Appendix: Figure B7)*, with residuals showing approximate normality and no influential outliers. We corrected minor heteroskedasticity by using Newey-West standard errors and bootstrapping.

In short, our analysis reveals a challenging reality which is that Pakistan's growing working-age population has not translated into greater capital formation, which means that the expected "demographic dividend" through capital formation channel has failed to materialize. This actually suggests that without supportive policies to raise savings, incentivize investment over consumption and strengthen the financial sector, only a larger workforce cannot drive the capital

formation which is needed for growth. While capital is still crucial for growth, as suggested by our Path b result, demographic change alone is not driving it.

Our finding differs from Jehan and Khan's results, which found a link between demography and capital over the long run (1960–2015), mostly because our focus was on the post-1990 period, which suggests that this link has broken down during the very years when Pakistan's demographic dividend was most accessible. Also, this could be because we tested only the capital pathway. The slight direct effect of the working-age population on GDP suggests that other channels like human capital or productivity might still be active and worth exploring in this case. For now, however, the evidence is clear that Pakistan's demographic opportunity has not translated into the investment driven boom required for a country's sustained growth.

5.3 Forecasting

5.3.1 VAR forecasting

The stability plot confirms that VAR is dynamically stable as all roots lie within the unit circle. Looking at the impulse response function to a shock to fertility, changes in working age population and dependency ratios are negligible as is characteristic of demographic variables which are not captured in the short-run VAR framework (See Appendix: Figure B8).

For domestic savings and capital formation, responses are muted to fertility shocks, indicating that the second dividend may only be potentially realised over larger horizons and not in the short-run. GDP per capita shows a noticeable response but converges to zero eventually. We interpret this, not as contradiction to the theory, rather that demographic transition gains may materialise through in the long-run.

FEVD decomposes error variance of GDP per capita into its own shocks as well as those from other variables (*Appendix: Figure A9*). Own shocks dominate, explaining the majority of the variance while savings and capital formation also play a crucial role. Demographic variables have a lower share but still impact error variance, nonetheless, supporting a mediated demographic dividend mechanism. The VAR forecasts above capture the duration of Pakistan's demographic transition window, in which the benefits of a decline in fertility may be reaped.

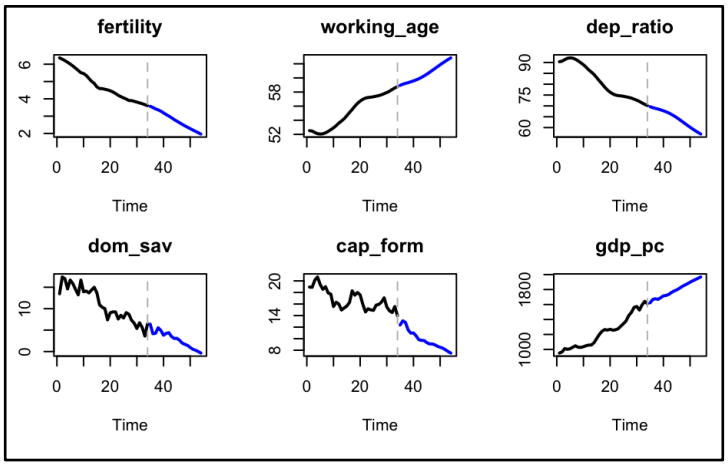


Figure 2: VAR Forecast Results in levels (time period: 1990-2043).

The level forecast derived from differenced VAR suggests continuous decline in the fertility rate and dependency ratio while working-age population gradually increases. As working age population increases, its rate of change decreases, consistent with available literature that Pakistan is now in the latter half of its demographic transition (Dur-e-Nayaab 2006).

Domestic saving and capital formation suggest volatile dynamics. Based on predictions, we infer that savings growth does not accelerate proportionally as the working age population increases (despite a predicted decline in the dependency ratio as well). Capital formation is sensitive to past values and exhibits a volatile, declining trend. Forecasts for GDP per capita contain an upward trend, consistent with historical patterns.

Projections based on this model are based on past relationships and therefore do not account for any major shocks that may occur. These findings are also consistent with the literature, which suggests that demographic dividends are conditional on the policies implemented, such as human capital investment and labor market absorption.

5.3.2 VECM Forecasting

Figure 3 shows the VECM forecasts for GDP per capita, domestic savings, and capital formation, with the dashed line marking the start of the forecast period. GDP per capita follows a steady upward path, reflecting the presence of a stable long-run relationship among the endogenous variables. This suggests that income growth can be sustained during the demographic transition period. In contrast, the domestic savings forecast declines persistently and turns negative towards the end. Capital formation remains relatively stable but does not exhibit a sustained increase (Table A5 in the appendix shows some forecast values).

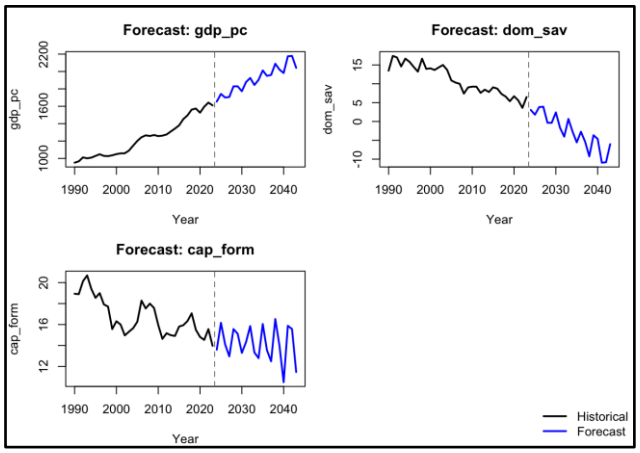


Figure 3: VECM Forecast Results

6 Conclusion

In this paper we investigate the application of demographic dividends in the context of Pakistan, offering insights into how Pakistan's declining fertility rates and rising working age population have shaped its macroeconomic dynamics of savings, investment and growth. We investigated this using time-series models, namely VECM to study long-run relationships, a Mediation Model to test transmission channels and VAR-based forecasting to project future trends. We find evidence that Pakistan is not benefiting from demographic dividends of a fertility decline, and subsequent increase in the working age population.

Despite our findings, we acknowledge the limitations of our research, which mainly stem from sample size problems. Owing to Pakistan's late entry into the fertility decline stage, post-1990, we were left with fewer observations than ideal for a robust time series analysis. However, our study still builds on current literature and examines how demographic changes interact with Pakistan's economic structure.

Pakistan must convert the opportunity of a working age population increase, into actual gains for the economy in the form of savings translating into capital formation and economic growth. This paper flags for policymakers that despite entering the demographic transition phase, Pakistan has not been able to reap benefits, and therefore policy action must be taken to foster financial development, incentivize investment and fully utilize the potential of its growing workforce to convert Pakistan's demographic transition into a demographic dividend.

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Appendix

Appendix A:

Variable	ADF Statistic	p-value	Stationarity (5%)
Fertility rate	−1.034	0.918	Non-stationary
Domestic savings	−3.329	0.084	Non-stationary
Capital formation	−2.409	0.415	Non-stationary
Working-age population	−5.143	0.010	Stationary
Dependency ratio	−4.887	0.010	Stationary
GDP per capita	−2.194	0.498	Non-stationary

Table A1: ADF results for Variables.

Note: Stationary assessed at the 5% significance level.

Variable	Vector 1	Vector 2	Vector 3
gdp_pc	1.00000	1.0000	1.00000
dom_sav	207.9270	179.0645	40.61242
cap_form	19.6946	392.1336	26.51998
Constant	-18307.51	-25145.50	-3618.334

Table A2: Normalized Long-Run Cointegrating Equation (β)

Equation	Vector 1	Vector 2	Vector 3
Δgdp_pc	0.09833	-0.01823	-0.00899
Δdom_sav	-0.00608	-0.00055	0.000033
Δcap_form	0.00202	-0.00030	-0.00263

Table A3: Speed of Adjustment Coefficients (α)

Dependent Variable	ECT (α)	Working Age	ΔGDP pc (t−1)	ΔDomestic Savings (t−1)	ΔCapital Formation (t−1)	Adj. R^2
ΔGDP per capita	0.0983**	26.021**	−0.103	15.414***	7.185	0.4588
	(0.035)	(9.068)	(0.223)	(4.766)	(5.807)	
ΔDomestic Savings	−0.0061***	−1.584***	−0.012	−1.086***	0.019	0.5478
	(0.001)	(0.371)	(0.009)	(0.195)	(0.238)	
ΔCapital Formation	0.0020	0.517	0.010	0.151	−0.231	0.1452
	(0.001)	(0.318)	(0.008)	(0.167)	(0.203)	

Table A4: Speed of Adjustment Coefficients (α)

Year	fertility	working_age	gdp_pc	dom_sav	cap_form
2024	3.621	58.934	1652.233	3.734	13.556
2029	3.631	59.753	1839.146	-0.203	15.852
2034	3.620	60.720	1914.738	-2.976	13.566
2039	3.567	61.702	2034.376	-5.898	13.631

Table A5: Forecasted Values Using VECM

Appendix B

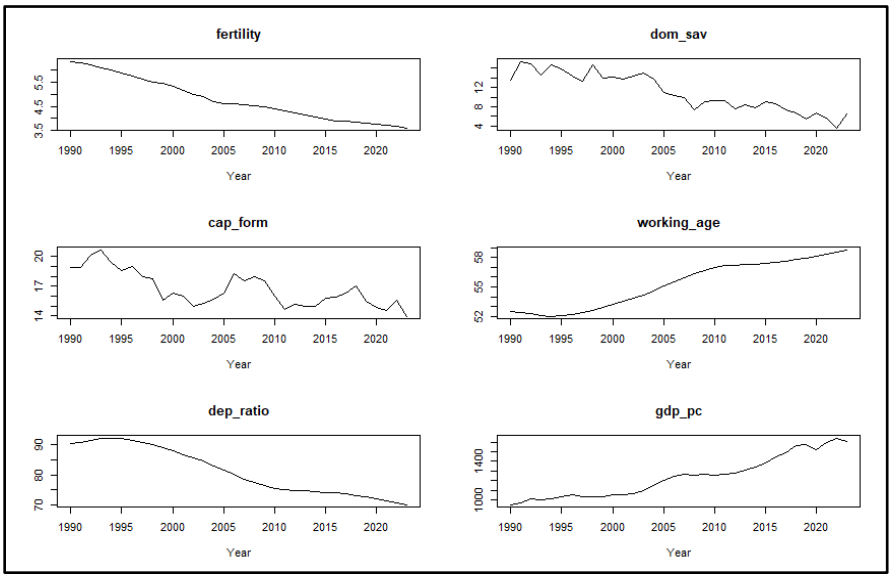


Figure B1: Autoplots for all time series variables

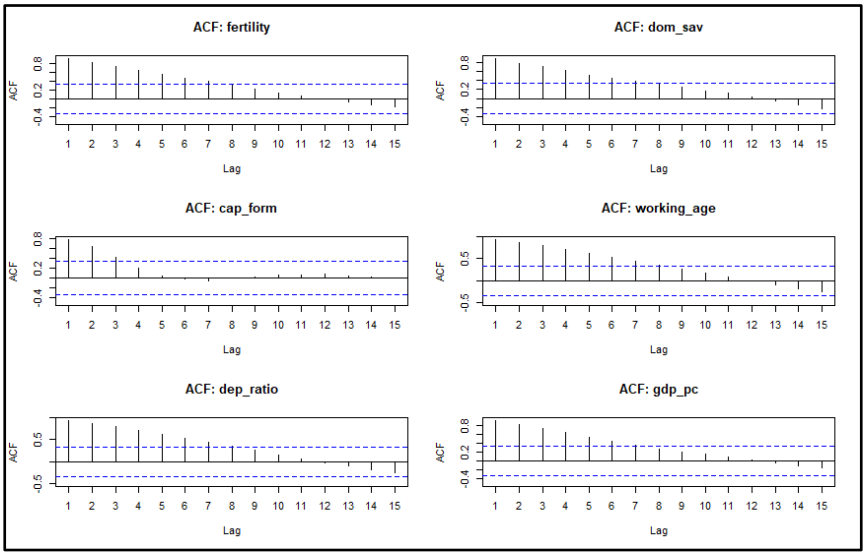


Figure B2: ACF plots for all variables

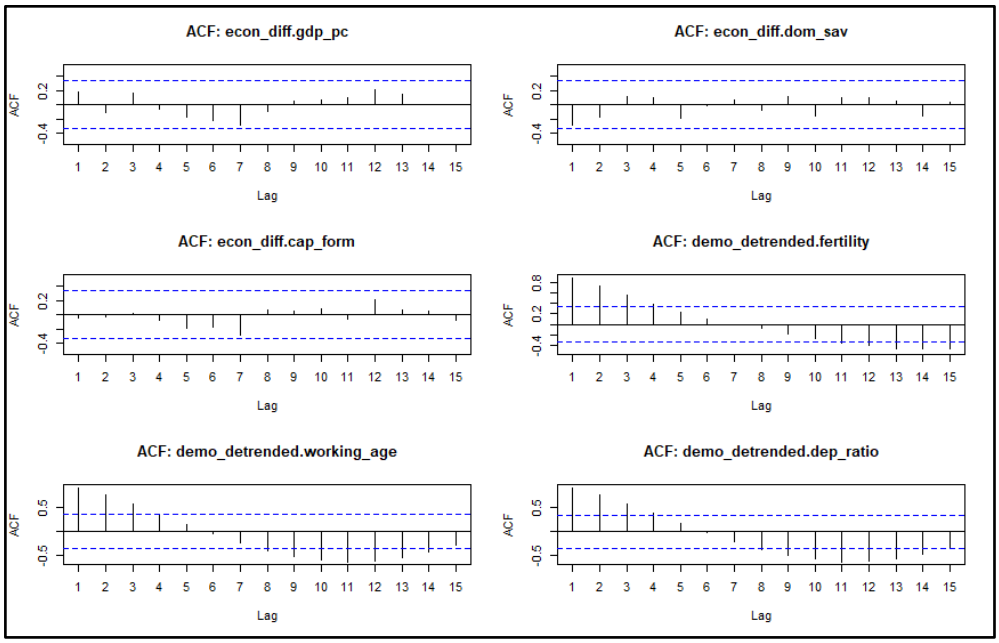


Figure B3:ACF of all processed series

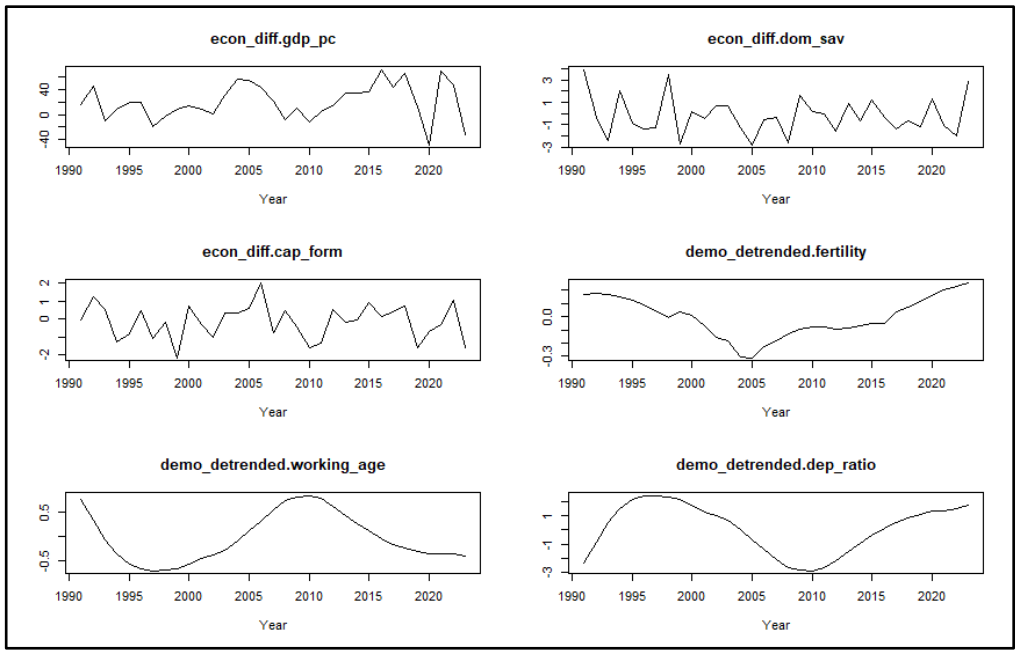


Figure B4: Autoplots of processed series

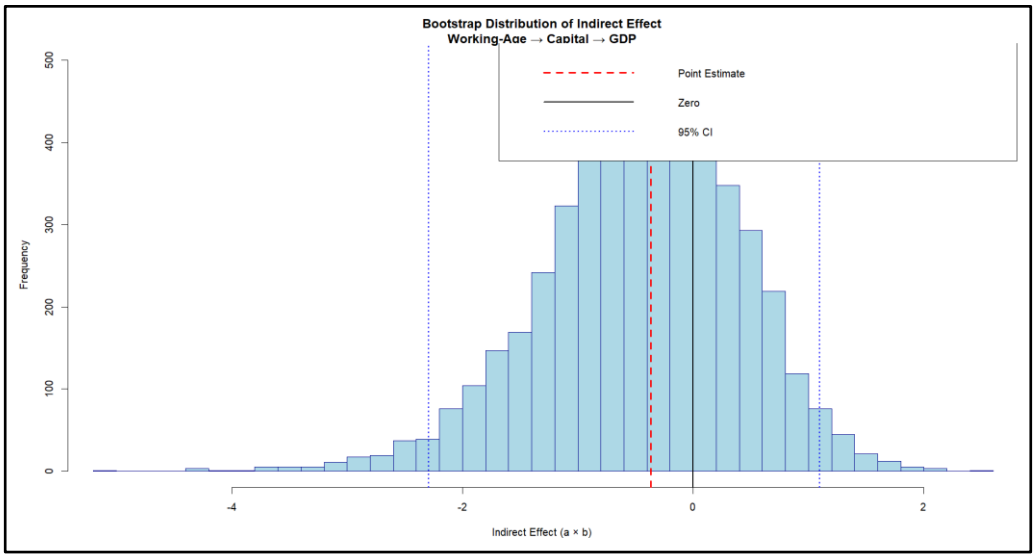


Figure B5: Bootstrap Distribution of Indirect Effect

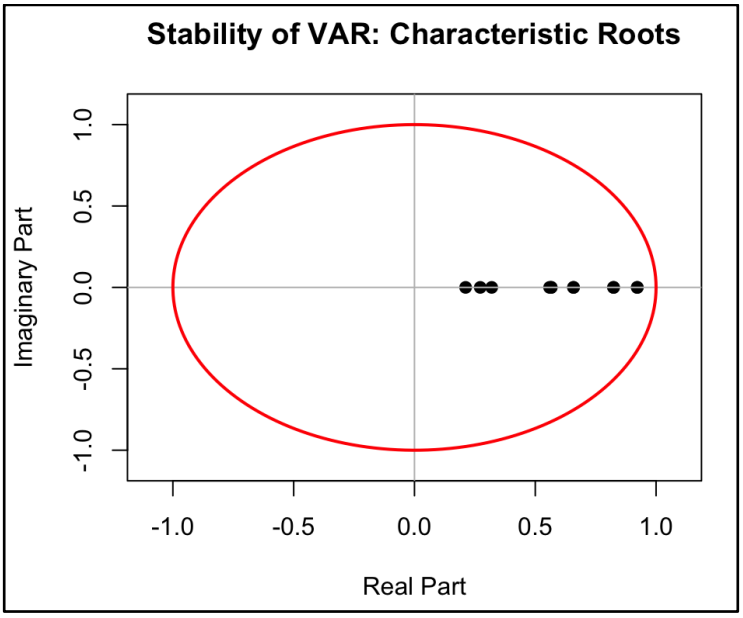


Figure B6: Stability Test for VAR Model

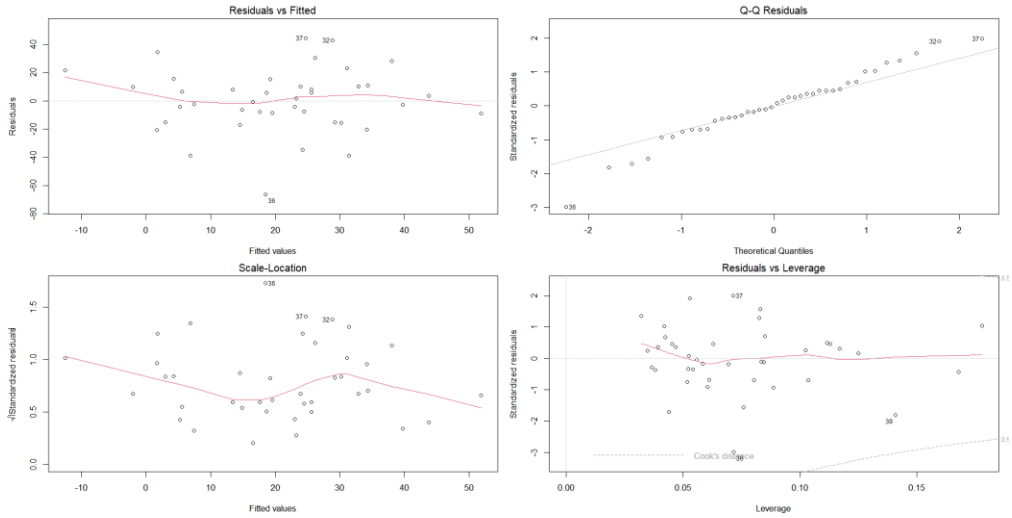


Figure B7: Diagnostic Checks (Mediation Model)

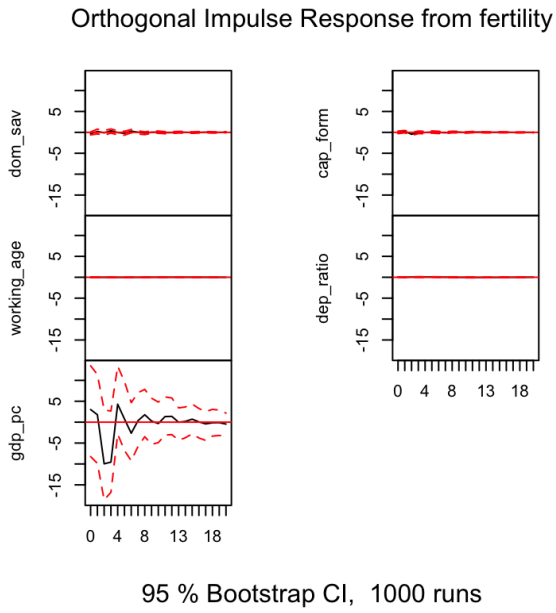


Figure B8: Impulse Response Function to a shock to fertility

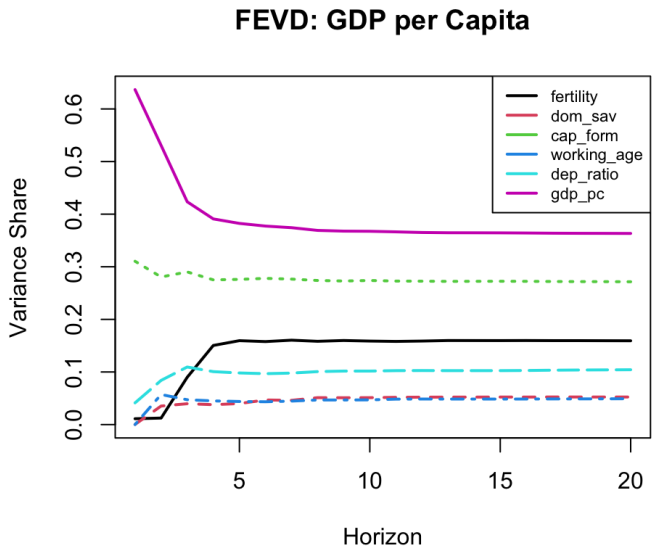


Figure B9: Forecast Error Variance Decomposition of GDP per capita